

FUNDAMENTALS OF ELECTRICAL AND ELECTRONICS ENGINEERING
(NES-102/202)

Module No.	Title of the Module	List of Topics
Unit I	Electrical Circuit Analysis	Voltage and current sources, dependent and independent sources, source conversion, DC circuit's analysis using mesh & nodal method, Thevenin's & superposition theorem, star-delta transformation. 1-phase AC circuits under sinusoidal steady-state, active, reactive, and apparent power, physical meaning of reactive power, power factor, 3-phase balanced and unbalanced supply, star and delta connections.
Unit II	Electrical Machines (Static & Dynamic)	Transformers: Magnetic Circuits: Review of laws of electromagnetism, Flux, MMF and their relation, analysis of the magnetic and electric circuit. Single-phase transformer: Basic concepts, constructional features, EMF equation, voltage, current, and impedance transformation, Equivalent circuits. Electrical Machines: DC Machines: Constructional features, working principle, EMF equation, types of DC machines, and their characteristics. Induction Machines: Constructional features, working principle, EMF equation, the concept of slip and torque-slip characteristics. Synchronous Machines: Constructional features, working principle, and EMF equation.
Unit III	Semiconductors	Conductivity of insulators, metals, and semiconductors in terms of energy bands, the chemical bond in Si and Ge, conductivity of intrinsic semiconductors, extrinsic semiconductors: n-type and p-type semiconductors, Hall Effect in semiconductors, Mechanism of current flow: drift and diffusion, Einstein relation, semiconductor materials: Element semiconductor, II-VI compound, III-V compounds, ternary and quaternary compounds. V-I characteristics of PN-junction diode. Diode equivalent circuit, diode as a switch, diode testing.
Unit IV	Diode Applications and Bipolar Junction Transistor	Rectifiers: Half wave, center tapped and bridge full-wave, Zener diode regulator and voltage multiplier, clipping and clamping circuits. Transistor: Introduction, classification (CC, CB & CE), configurations, transistor as an amplifier, testing of transistor, load line analysis, biasing of the transistor, bias compensation, transistor as a switch.
Unit V	Field Effect Transistor & Introduction of Digital Electronics	Construction and characteristics of JFET, JFET Biasing circuits, JFET Amplifier, MOSFET construction, and characteristics. Logic Simplification and Combinational Logic Design: Review of Boolean Algebra and De Morgan's Theorem, SOP & POS forms, Canonical forms, Karnaugh maps up to 6 variables, Binary codes, Code Conversion.